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Notiser — Notes

Palaeomagnetism of a probable meteorite impact, the Dellen structure

GÖRAN BYLUND

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Twenty orientated AF-demagnetized samples of dellenite from the Dellen depression, which is a possible meteorite impact site, rendered a virtual geomagnetic pole position at 68°N., 165°E. This pole position agrees roughly with Lower Tertiary palaeomagnetic poles determined for stable Europe.

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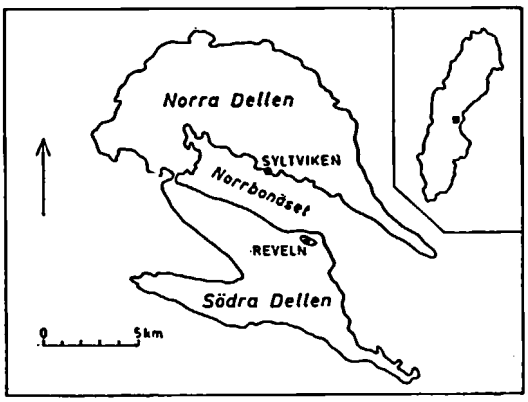


Fig. 1. Map of the Dellen Lakes area with sampling sites.

The Dellen Lakes are situated at 61.92°N., 16.53°E. and occupy a rounded depression, which has been suggested to be one of the possible meteorite impact sites in Sweden (Fredriksson & Wickman 1963, Svensson 1968). Some investigators, however, hold that the structure is of volcanic origin (Redaelli 1957, Lundegårdh 1967). Dellen is the only Swedish structure of this type where there are outcrops of solid rocks (dellenite) formed during the event, and where it is thus possible to obtain orientated samples. Ten orientated samples were collected from each of the two available exposures of dellenite, and the natural remanent magnetism (NRM) was studied. These two sites are on the island of Reveln and on the northern shore of the Norrbonäset peninsula, at Syltviken (Map. in Fig. 1). Precambrian gneisses and granodiorites make up the bedrock surrounding the dellenite area.

The composition of dellenite is intermediate between andesite and rhyolite. There are three varieties: 1. tuffs and tuffbreccias, 2. devitrified andesite, and 3. glassy andesite (Redaelli 1957). The samples taken consist of devitrified andesite. This rock is made up of phenocrysts of plagioclase and pyroxene set in a groundmass of alkali feldspar, quartz, and unidentified dark silicates. There is some magnetite which occurs as idiomorphic phenocrysts or forms microlites in the rock matrix.

Sampling was done with a small portable core drill and an orientation device of the kind commonly used in palaeomagnetic work (Doell & Cox 1967, Hesley 1967). The core azimuth was measured with a magnetic compass. For measuring a specimen of 2.3 cm length was cut from each core. The core diameter is 2.5 cm.

A four-probe fluxgate magnetometer (Förster Oerstedmeter) was used for measuring the NRM. The susceptibility was measured with a Soiltest MS-3 susceptibility bridge. The alternating field (AF) technique was applied to remove the soft viscous magnetic components.

Pilot specimens from each site were demagnetized stepwise in alternating fields up to a peak value of 500 Oe. The steps were 50 Oe. Demagnetization diagrams for the pilot specimens are given in Fig. 2. They show that, during demagnetization from the 100–150 Oe level and upwards, the magnetization vector decreases along a straight line towards the origin. This implies that only a hard stable magnetism remained at that level. The agreement of the curves with straight

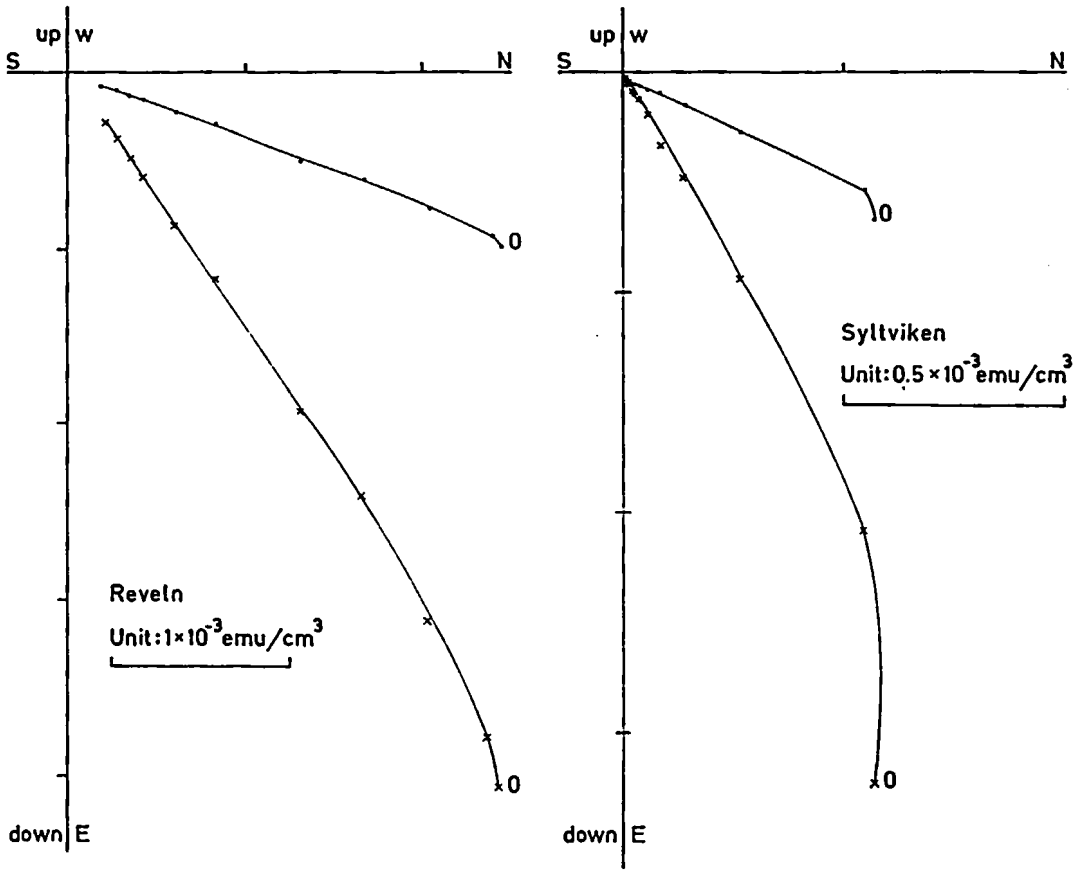


Fig. 2. Pilote specimen demagnetization diagrams. The plotted points represent successive positions of the end of the magnetization vector during progressive demagnetization. Points: projections on the horizontal plane; crosses: projections on the N./S. vertical plane. Orthogonal projection is used.

lines was tested by a least-square approximation procedure. It was concluded that the soft viscous components present initially were removed at the 300 Oe level, and all the remaining specimens were demagnetized stepwise to that level. The NRM directions of each specimen after AF-demagnetization to the 300 Oe peak value are plotted in Fig. 3.

The average initial NRM was $3 \cdot 10^{-3}$ emu/cm³ for the Reveln samples, and $2 \cdot 10^{-3}$ emu/cm³ for those from the Syltviken. The average susceptibility was $4 \cdot 10^{-4}$ emu/cm³ and $3 \cdot 10^{-4}$ emu/cm³ respectively. This gives mean Q-values of 15 and 13 for the two sites. The Q-value is the ratio of the remanent magnetization to the magnetization induced in the sample by the Earth's field at the sampling locality (0.5 Oe). Mean magnetic directions found after AF-demagnetization are listed in Table 1. Each specimen was given unit weight in the calculations, which were made according to Fisher (1953).

The consistency of directions during demagnetization, the high Q-value and the absence of

Table 1. Mean directions of magnetization after demagnetization to 300 Oe peak field.

Site	*N*	*D*	*I*	*k*	α_{95}
Reveln	10	17.5°	62°	97	5°
Syltviken	10	13°	60°	227	3°
Combined	20	15°	61°	135	2.8°

N: Number of specimens measured.
D: Declination, measured clockwise from true north.
I: Inclination, measured from the horizontal.
k: Best estimate of Fisher's precision parameter.
α_{95}: Half-angle of the cone of confidence at the 95 % level.

Table 2. Tertiary palaeomagnetic data for Europe.

Formation or area	Age	North pole position		Reference
Basaltic flows of the Faeroe islands	50–60 m.y.	77°N.	161°E.	Tarling 1970
British results *a)*	Tertiary	77°N.	171°E.	Tarling 1970
Antrim basalt flows, Northern Ireland	Tertiary	69.5°N.	163°E.	Wilson 1970
W. Europe and Russian Platform, mean	Upper Tertiary	79°N.	180°E.	McElhinny 1973
W. Europe and Russian Platform, mean	Lower Tertiary	75°N.	164°E.	McElhinny 1973
Dellenite, Dellen Sweden		68°N.	165°E.	This paper

a) British results from 16 mean pole positions of variable reliability, compiled by Tarling in 1970.

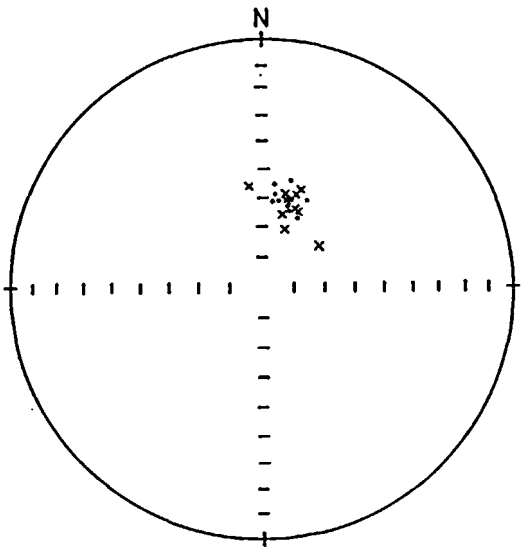


Fig. 3. Magnetic directions of all measured specimens after demagnetization to a 300 Oe peak field value. Reveln: crosses, Syltviken: points. Equal area stereographic projection, lower hemisphere, is used.

chemical alterations in the magnetite imply that the stable remanence observed is the original thermo-remanent magnetization (TRM) acquired during the formation of the rocks in the ambient geomagnetic field.

The virtual geomagnetic pole calculated from the Dellen data is situated at 68°N., 165°E. For comparison, available magnetic poles from Tertiary igneous rocks of stable Europe are listed in Table 2. The Dellen pole is situated in the vicinity of the listed Lower Tertiary poles, thus implying that the dellenite dates from this period. Previously, Svenonius (1887) had suggested a Tertiary age of dellenite formation. In discussing the magnetic pole position obtained from the dellenite it must be remembered that geologically speaking, the event — either meteoritic or volcanic — which created the Dellen structure and the dellenite comprised a short period. Moreover we must take into account that sampling was possible at two sites only and consequently the palaeosecular variation was not averaged out. The range of directional changes which are due to secular variation may be about 10 to 20 degrees to either side of the average pole position (Irving 1964). This uncertainty must be considered when comparing the Dellen pole with established palaeomagnetic poles. This study thus presents a "spot reading" of the ancient geomagnetic field, giving a result which in general terms confirms Svenonius' opinion of a Tertiary dellenite age.

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New names for ostracode genera

ROGER SCHALLREUTER

Schallreuter, R., 1974: New names for ostracode genera. *Geologiska Föreningens i Stockholm Förhandlingar,* Vol. 96 p. 278. Stockholm, September 30, 1974.

The generic names *Severella* Schallreuter, 1964, *Altha* Neckaja, 1958 and *Asymmetrella* Neckaja, 1973 (Ostracoda) are preoccupied by *Severella* Sapelnikov, 1963 (Brachiopoda), *Altha* Walker, 1862 (Insecta) and *Asymmetrella* Cobb, 1920 (Vermes). The new names for the ostracode genera will be *Severobolbina, Neckajatia* and *Allertemmysa* respectively.

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According to an information of Dr. Valdar Jaanusson, Stockholm, the name of the ostracode genus *Severella* Schallreuter, 1964 is preoccupied by the name of the brachiopod genus *Severella* Sapelnikov, 1963. The new name *Severobolbina* nom.n. is introduced here to substitute *Severella* Schallreuter, 1964 (type species *Strepula elliptica* Steusloff, 1895).

Another preoccupied ostracode generic name is *Altha* Neckaja, 1958. In a letter of March 2nd, 1967, the author informed Dr. Neckaja that her genus is a junior homonym of *Altha* Walker, 1862 (Insects). Neckaja never published a new name for her genus. Therefore the name *Altha* Neckaja, 1958 is replaced here by the new name *Neckajatia* nom.n. (type species *Altha modesta* Neckaja, 1958) in honour of the meritorious ostracode worker.

A further preoccupied ostracode generic name of Neckaja is *Asymmetrella* Neckaja, 1973. It is a junior homonym of *Asymmetrella* Cobb, 1920 (Vermes). The name *Allertemmysa* nom.n. (= anagram of *Asymmetrella)* is proposed to substitute *Asymmetrella* Neckaja, 1973 (type species *Asymmetrella prenaiensis* Neckaja, 1973).

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